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Drag at 21.0 m s^{-1} : $f_D = \frac{110}{21} = 5.24 \text{ kN}$

Drag at 15.0 m s^{-1} : $\frac{f_D'}{f_D} = \left(\frac{15.0}{21.0}\right)^2 \Rightarrow f_D' = 2.67 \text{ kN}$

Force of engine at 15.0 m s^{-1} : $F_{\text{engine}} = \frac{110}{15} = 7.33 \text{ kN}$

Hence resultant force = $7.33 - 2.67 = 4.66 \text{ kN}$